LYU JIAXIN

Guangzhou, China | 18224734143@163.com | +86 18224734143
linkedin.com/in/jiaxin-lyu81 | github.com/jiaxinlyu-a11y | jiaxinlyu-a11y.github.io

SUMMARY

MSc Statistical Data Science student (UCD) seeking biostatistics and statistical analysis roles. Experience in regression modelling and model selection, survival analysis, simulation study design and sensitivity analysis in R. Re-analysed a retracted imaging study for my thesis, designed a survival-analysis simulation study, and built an AI-assisted data-recovery tool whose accuracy I tested.

EDUCATION

University College Dublin (UCD), MSc Statistical Data Science (GPA 3.54/4.0) *Sep 2025 – Present*
Courses: Survival Models, Mathematical Statistics, MCMC, Machine Learning, Advanced Predictive Analytics, Bioinformatics

Harbin University of Commerce, BSc Statistics *Sep 2019 – Jun 2023*

SKILLS

Statistics: linear regression, model selection (AIC), sensitivity analysis and diagnostics (HC3, robust regression), survival analysis (Cox, log-rank, RMST), simulation study design (ADEMP), machine learning (CNN, DNN)

Programming: R (survival, Keras), Python (OpenCV, Streamlit), SQL, SAS

Tools: Git/GitHub, Quarto, Markdown, Excel; AI-assisted development (I set requirements and validate)

Bioinformatics: NCBI protein BLAST, Clustal Omega alignment and phylogenetic trees; alignment and tree-building algorithms (Needleman-Wunsch and Smith-Waterman implemented in R; BLAST, WPGMA, neighbour joining); GenBank, UniProt, PDB

Languages: English (IELTS 6.5 overall), Mandarin (native)

PROJECTS

KETO-CTA Re-Analysis | MSc thesis, sole analyst | R, Python

- **Goal:** using the released data of a retracted imaging study (100 participants, 2 CT scans), test whether baseline plaque and ApoB predict later plaque.
- **My role:** sole analyst. Rebuilt the published figures to check the data reproduced them; as predictors overlap, used joint models and compared 3 regression models (AIC, backward elimination); chose codings from each variable's distribution (log1p, sqrt, binary); ran sensitivity analyses on two outcome definitions and diagnostics (HC3, robust regression).
- **Data recovery (AI-assisted tool):** participant-level ApoB was never released. I set the requirements, had AI write the Python tool (OpenCV, Streamlit) and tested it against known values: on 2 plots of 100 points, 100 and 99 were matched (94 and 87 found automatically, the rest added by hand); mean error 0.13–0.25 mm^3, largest 1.07 mm^3 on an axis of about 160 mm^3 (about 0.7%).

Survival Analysis Simulation | Personal project | R, Quarto

- **Goal:** compare how Cox, log-rank and RMST tests perform under different conditions.
- **My role:** designed (ADEMP), coded in R and reported in Quarto on my own: two groups with the hazard ratio switching at month 6; 4 effect patterns × 2 censoring levels (about 20%, 50%) × 3 sample sizes (100, 300, 600 per group) = 24 scenarios, 1,000 repetitions each; estimand: 24-month RMST difference.
- **Result:** judged by rejection rate, bias and Monte Carlo standard error, and verified the simulation first: Type I error 0.043–0.066 (target 0.05); bias under twice the Monte Carlo standard error in all 24 scenarios. Code and report are public.

Check-up Planner | AI-assisted tool | JavaScript, bilingual, runs in the browser

- **Goal:** turn published screening guidelines into check-up advice that adapts to the person, with evidence level and source for each item.
- **My role:** selected 41 rules and fixed their sources (USPSTF, ACS, ACG, Chinese guidelines, OSHA, GBZ 188-2025); designed adjustments for lifestyle, personal and family history where guidelines are silent; tested repeatedly; had AI write the code.
- **Result:** five evidence levels; 9 tests guidelines advise against for people without symptoms are listed separately.

Human Activity Recognition | UCD coursework | R, Keras. Compared a DNN and a CNN on wearable-sensor time series (19 activity classes); the CNN generalised better, with 97.8% test accuracy.

EXPERIENCE

Chengdu VisionVS Technology Co., Ltd., Product Assistant & Delivery Engineer, Chengdu, China *Jul 2023 – Mar 2025*

- Wrote requirements, design specifications and technical documentation with clients for interactive electronic technical manuals (IETM, IETP) and an EHS management system; led an internal project-management system from a survey of four tools to delivery.
- Benchmarked five vendors (about 196 core functions) and compared three EHS systems function by function to define a standard product; trained users and worked with developers on iterations.